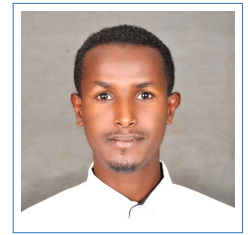


Habtamu Molla

Curriculum Vitae

Department of IT
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✉ habtamumullu@gmail.com
ResearcherID [in](#) [Linkedin](#)
[ResearchGate](#) [GitHub](#) [ORCID](#)
Portfolio link: [Website](#)



Education

- 2021-2023 : **MSc. in Computer Networks and Communications, Wollo University, KIoT, Ethiopia,**
Thesis Title: Design a Robust DDoS Attack Detection and Mitigation Scheme in SDN-Edge-IoT by Leveraging Machine Learning.
Grade: CGPA: 4/4, with thesis result Very Good
- 2024-2025 **Higher Diploma in Teaching in Higher Education (HDP), Debark University, Ethiopia.**
Grade: Successfully Completed (Awarded June 20, 2025)
- 2016-2019 : **BSc. in Information Technology (IT), Bule Hora University, Ethiopia.**
Project Title: Android Mobile-based Student Instructor Parent Interaction System
Grade: CGPA: 3.91/4, with Project result A

Working Experience

Software Industry Experience

- 07/07/2026 – Present **Full Stack Developer, Walia Nexus PLC, Addis Ababa, Ethiopia,** Develop and maintain full stack software solutions across backend and frontend, design and implement APIs, manage database architecture, conduct testing and deployment support, produce technical documentation, and contribute to AI, data, and digital transformation projects..
- 18/06/2026 – Present **Individual Consultant (ICSMIS Dashboard), FCSC & UNDP, Addis Ababa, Ethiopia,** Designed and developed an enterprise executive dashboard for ICSMIS, a UNDP-funded initiative. Led full-stack development, integrated diverse modules via REST APIs, and delivered intuitive data visualizations and KPIs for executive decision-making. Managed the full SDLC from requirements to deployment and post-launch support..
- 17/04/2026 – 22/07/2026 **Software Business Analyst, Tria Trading PLC, Addis Ababa, Ethiopia,** Lead business requirements analysis, translate needs into functional specifications, collaborate with cross-functional teams throughout the SDLC, define user stories and acceptance criteria, coordinate UAT and defect resolution, support system integration and production, and drive process optimization and digital transformation initiatives..
- 20/02/2026 – 16/04/2026 **Data Scientist, Condigital Technologies PLC, Addis Ababa, Ethiopia,** Developed high-quality human-authored code for training and evaluating code-generative AI models using the PostWork platform and Google Colab. Built AI-powered BI solutions, engineered data pipelines, developed machine learning models, and applied Python, and AI frameworks to generate actionable insights and support data-driven decision-making..

Teaching Experience

- 08/02/2026 – Present **Lecturer in college of Computing and Digitalization under Information Technology (IT) Department, Ethiopian Public Service University, Addis Abeba, Ethiopia,** Responsible for teaching a wide range of courses outlined in the curriculum of Information Technology, Computer science, Computer Networks and Communications.

08/10/2019 – **Lecturer in Information Technology (IT), Debark University, Debark, Ethiopia**, Responsible
7/2/2026 for teaching a wide range of courses outlined in the curriculum of Information Technology,
Computer science, Computer Networks and Communications.

Project Awards & Recognition

- 2025 **Certificate of Digital Transformation (Community Engagement):** Awarded by Debark University and North Gondar communication Department on May 8, 2025, in recognition of my valuable contribution in developing, hosting a website for the North Gondar Zone Communication Department. This certificate acknowledges the significant support my work provided in enhancing communication services in the region. *Signed by the Technology Transfer and Community Service Directorate, V/President for Academic, Research, Technology Transfer & Community Service , and North Gondar Zone Communication Department Director. This website can be access via: <https://northgondarcom.gov.et>.*
- 2019 **Certificate of Application:** This is in recognition of my excellent effort and achievements in being an outstanding student and competent from our college in final year project entitled as **"Android Mobile-based Student Instructor Parent interaction system"** and one of the top three competent student form our college, which was given by Bule Hora university, Ethiopia.

Publications

Journal Articles

- 2025 Habtamu Hunegnaw Limenih, Abebe Belay Adege, Abrham Yaregal Alene, Habtamu Tariku Demasu, and Habtamu Molla Belachew. Automatic identification of amharic text idiomatic expressions using a deep learning approach. *IEEE Access*. IEEE, 2025.
- 2025 Mulatu Yirga Beyene, Salahadin Seid Musa, Habtamu Molla Belachew, Behaylu Tadele Alemu, Amogne Andualem Ayalew, and Melaku Lake Tegegne. Improving the Efficiency of a Message Validation System in a Vehicular Ad Hoc Network. *Journal of Computer Networks and Communications*, volume 2025, pages Article ID 8771020, 16 pages. Wiley, 2025.
- 2025 Habtamu Molla Belachew, Mulatu Yirga Beyene, Abinet Bizuayehu Desta, Behaylu Tadele Alemu, Salahadin Seid Musa, and Alemu Jorgi Muhammed. Design a Robust DDoS Attack Detection and Mitigation Scheme in SDN-Edge-IoT by Leveraging Machine Learning. *IEEE Access*, volume 13, pages 10194–10214, 2025.
- 2024 Behaylu Tadele Alemu, Alemu Jorgi Muhammed, Habtamu Molla Belachew, and Mulatu Yirga Beyene. A Comprehensive Detection and Mitigation Mechanism to Protect SD-IoV Systems Against Controller-Targeted DDoS Attacks. 2024.

In Conference Proceedings

- 2024 Abinet Bizuayehu Desta, Abebe Alemu Abetw, Kead Nigussie Werku, Nega Agmas Asfaw, Habtamu Molla Belachew, and Wubie Belay Alemu. Sub Event Relation Classification for Amharic Texts Using Deep Learning Approach. In *2024 International Conference on Information and Communication Technology for Development for Africa (ICT4DA)*, pages 205–210, 2024.

Project Experience

- Enterprise System **SmartOffice** (<https://smartoffice.moa.gov.et/>) – Worked as a **Software Business Analyst** at **Tria Trading PLC**. I led requirements engineering, business process analysis, stakeholder communication, system documentation, and user acceptance support for the SmartOffice enterprise platform. Contributed to the successful deployment of Record Management, Document Management, and Performance Management Systems across major Ethiopian public institutions, including Addis Ababa City Administration, the Ministry of Agriculture, and Ethio-Djibouti Railway Services.

Website	North Gondar Zone Government Communication Website (https://northgondarcom.gov.et/), I contributed as a full-stack developer, handled deployment and hosting via Plesk Control Panel, and continue to serve as system administrator responsible for ongoing maintenance, updates, and technical support to ensure reliable communication between the government and citizens. Completed on January 17, 2025, Debark, Ethiopia.
Mobile app	Android Mobile Based Student-Instructor-Parent Interaction System for Bule Hora University using Android, Java, MYSQL, and Firebase
Web system	Bule Hora web based clinic management system, and Bule Hora online registrar database management system using PHP, HTML, CSS, JavaScript, and MYSQL
Desktop app	College management system, Student management system, and Scientific calculator using VB.net, MYSQL, and Java

Professional Roles & Contributions

2024-8/2/2026	Indigenous Knowledge (IK) Assessment and Documentation Committee: Serving as a committee member at Debark University, Debark, Ethiopia.
2025-2026	Academic Board Member, Global Open Share Publishing: Serving on the academic board of Global Open Share Publishing Pty Ltd. https://www.gospub.com/academic-board
2025-present	Editorial Board Member, Deep Science Publishing: Recognized as a member of the Editorial Board by Deep Science Publishing for contribution to the peer-review process and editorial responsibilities. Official certificate issued acknowledging participation and service. https://deepscienceresearch.com/dsr/index

Peer-Review Recognitions

All reviewer recognitions verified via Web of Science ResearcherID: <https://www.webofscience.com/wos/author/record/MIO-9111-2025>

2024	IJECE (Intl. J. Electrical & Computer Eng.) – Active Reviewer. http://ijece.iaescore.com/
2025	IEEE Internet of Things Journal – Active Reviewer. https://ieee-iotj.org/
2025	IEEE Access – Active Reviewer. https://ieeeaccess.ieee.org/
2025	IEEE OJCOMS (Open J. Communications Society) – Active Reviewer. https://ieee-ojcoms.org/
2025	Artificial Intelligent and Application (AIA) – Active Reviewer. https://ojs.bonviewpress.com/index.php/AIA/index
2025	The Journal of Engineering Research and Sciences (JENRS) – Active Reviewer. https://www.jenrs.com/
2025	ICECER 2025 (2nd Intl. Conf. Electrical & Computer Eng. Researches) – Active Reviewer. Antananarivo, Madagascar (Dec 2025). https://www.icecer.com/2025
2025	ICT4DA 2025 (7th Intl. Conf. ICT for Development for Africa) – Active Reviewer. Bahir Dar, Ethiopia (Nov 2025). https://ict4daconf.org
2025	ICECCME 2025 (5th Intl. Conf. Electrical, Computer, Comm. & Mechatronics Eng.) – Active Reviewer. Zanzibar, Tanzania (Oct 2025). http://www.iceccme.com/2025/
2025	ACDSA 2025 (Intl. Conf. Artificial Intelligence, Computer, Data Sciences & Applications) – Active Reviewer. Antalya, Turkiye (Aug 2025). http://www.acdsa.org

All reviewer recognitions verified via Web of Science ResearcherID: <https://www.webofscience.com/wos/author/record/MIO-9111-2025>

Training Experience

- 2026 **Cybersecurity Job Simulation** – Forage (Feb 2026). Designed phishing simulations and interpreted results to develop threat analysis and security awareness skills.
- 2019 **Entrepreneurship & Business Plan Development** – Deliveryology Unit, Bule Hora University (Jun 2019). Completed two-day training for graduating students.
- 2024 **Android Developer Fundamentals Nanodegree** – Udacity (Aug 2024). www.udacity.com/certificate/e/04148496-4cd5-11ef-b5ef-838a93829c33
- 2024 **Data Analysis Fundamentals Nanodegree** – Udacity (Aug 2024). <https://www.udacity.com/certificate/e/9d4dd036-66d9-11ef-bf26-731a3f1384bf>
- 2025 **Artificial Intelligence Fundamentals Nanodegree** – Udacity (Jan 2025). <https://www.udacity.com/certificate/e/9beff51e-c13a-11ef-bba9-c31472719a67>
- 2024 **Programming Fundamentals Nanodegree** – Udacity (Aug 2024). <https://www.udacity.com/certificate/e/aad101e4-6213-11ef-acce-eb0035983c7c>
- 2024 **Master Class for Teaching Online (Open edX)** – EdPlus | ASU Online (Sept 2024).
- 2025 **Paperpal Live Workshop** – "How to Simplify Your Literature Review Writing with Paperpal" by Dr. Sowmiya Rani, Cactus Communications (Mar 2025).
- 2024 **IEEE Authorship & Open Access Symposium** – IEEE (Oct 2024). Participated in tips and best practices session with IEEE Editors; certificate issued by Rachel Berrington, Director, IEEE Client Services.

Research Interests and Experience

My research interests and experience includes in securing the Internet of Things (IoT), Software-Defined Networking (SDN), Edge and Fog Computing, Machine Learning and Deep Learning for IoT, Cybersecurity in IoT, Vehicular Ad Hoc Networks (VANET) and Software-Defined Internet of Vehicles (SD-IoV), Natural Language Processing (NLP), Network Security, Cloud Computing, and Cyber-Physical Systems.

Programming Skills

Languages	Python, JavaScript, TypeScript, Java, Kotlin, Swift, C++, C, R, Rust
Web app	React, Angular, Vue.js, Node.js, Express, Django, Flask, PHP, HTML5, CSS3, SASS, GraphQL, RESTful APIs
Desktop App	Electron, JavaFX, .NET, WPF, Qt, GTK
Mobile App	Flutter, React Native, Swift, Kotlin, Java, Android SDK, Xcode, Firebase
AI Tools	TensorFlow, PyTorch, Keras, OpenCV, Scikit-learn, Hugging Face, spaCy, Pandas, NumPy
Databases	MySQL, PostgreSQL, MongoDB, Firebase, Neo4j, SQLite, Apache Cassandra, Redis

Language

Amharic	Fluent
English	Advanced
Geez	Intermediate

References

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